Research question:

What is the detection accuracy of the YOLO and ZED depth pipeline?

Objective

Quantify the 3D position error (x, y, z) of the YOLO and ZED depth pipeline on a stationary tennis ball across a known grid of positions and identify systematic biases to inform downstream use of the pipeline.

Setup

Equipment: ZED stereo camera, NEURAL depth mode. Ball resting on the ground plane, placed stationary at grid positions measured with a ruler from the left camera lens to the ball centre.

Coordinate system (camera frame):

- X - left/right (positive = right)
- Y - up/down (positive = down)
- Z - distance from camera

Camera tilt: pitch = -0.01°, roll = -0.09° (measured via ZED IMU). Effect at 0.80 m: <1.3 mm - negligible.

Grid: 45 captures total. Z in {0.57, 0.67, 0.77} m, X in {-0.7 … 0.7} m in 0.1 m steps, ball resting on the ground plane (Y = 0.28 m). All positions measured with a ruler, manual measurement accuracy is ±1 cm.

Camera intrinsics - fx and fy (focal lengths in pixels along the horizontal and vertical axes) and cx and cy (principal point coordinates, the pixel location of the optical axis) - are stored in `intrinsics.json` in the dataset directory, written automatically on first capture.

Detection Pipeline

1. **YOLO v8x** detects the ball and returns a bounding box (bbox).
2. **Front-face depth sampling**: within the bbox, the cluster of pixels within 0.5 cm of the minimum depth is selected. Their average pixel position is used as the (u, v) reference point and their median depth as Z.
3. **XYZ calculation**: X and Y are read from the pre-computed XYZ map at pixel (u, v), using the depth at that pixel: `x = (u - cx) * z_uv / fx`, `y = (v - cy) * z_uv / fy`. Z is the median depth of the front-face cluster. The two depth values (z_uv and median z) are nearly identical since all front-face pixels are within 0.5 cm of each other.

The front-face approach is used instead of the bbox centre to reduce Y error caused by background/floor pixels at the bottom of the bbox pulling the centroid down. Selecting only the closest pixels removes most of that effect. However, because the camera is mounted above the ball, the closest visible surface of the ball is its top rather than its centre, so the measured point is still above the true ball centre and a remaining Y error is expected (see Analysis).

Results

Detection rate: 41 / 45. The 4 missed captures (caps 30, 31, 44, 45) correspond to the ball being physically outside the camera's field of view at extreme lateral positions at the two shorter Z distances.

Summary (camera minus ruler):

Axis	Mean error	Std	Character
X	-1.1 cm	0.8 cm	Small, no strong systematic trend
Y	-5.0 cm	1.1 cm	Systematic negative bias
Z	4.1 cm	0.8 cm	Systematic positive bias
|Error|	6.7 cm	0.9 cm	-

Axis	Mean error	Std	Character

Plots - each figure shows one error metric vs true X position, split into three panels by Z distance. The dashed line is the per-panel mean; the gray band is the ±0.01 m ruler uncertainty. Colors indicate depth distance:

- Blue - Z = 0.57 m
- Orange - Z = 0.67 m
- Green - Z = 0.77 m

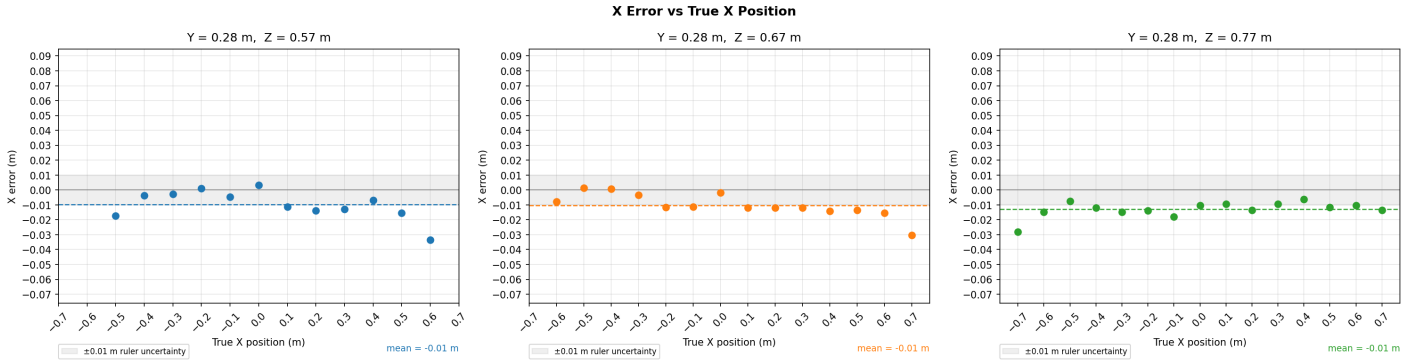


Figure 1: **X error vs true X position**: lateral (left/right) detection error. Values are scattered close to zero with no clear trend across X position or Z distance, indicating reliable lateral localisation.

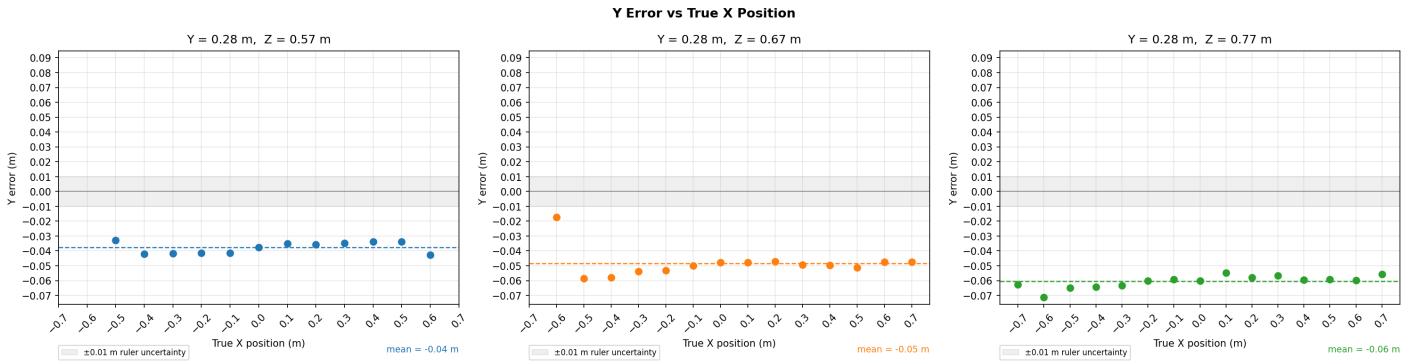


Figure 2: **Y error vs true X position**: vertical detection error. A strong consistent negative bias is visible across all X positions and Z distances, meaning the pipeline places the ball systematically higher than the true position.

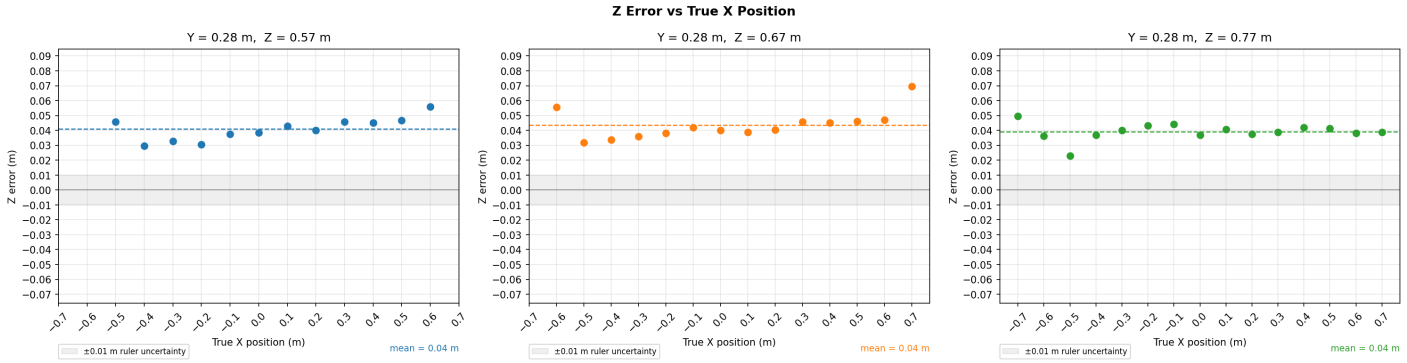


Figure 3: **Z error vs true X position**: depth detection error. A consistent positive bias appears at all positions with a slight increase at extreme lateral positions (ball near the left or right edge of the camera frame) where stereo matching quality decreases.

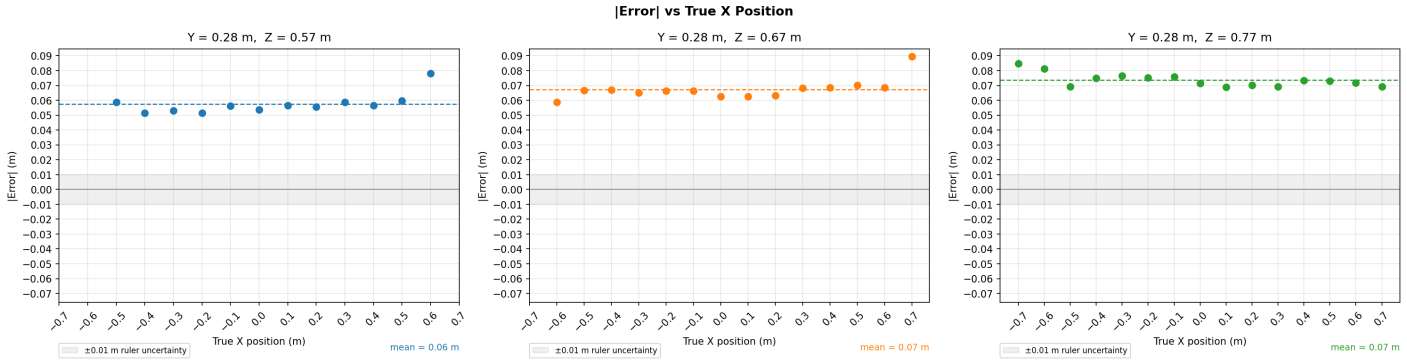


Figure 4: **Combined |Error| vs true X position**: 3D Euclidean distance between camera measurement and true position. The magnitude is dominated by the Y and Z systematic biases and their combined variation sets the overall accuracy floor.

Analysis

The pipeline shows reliable lateral (X) localisation and two consistent systematic biases: a negative Y error and a positive Z error, both present across all tested positions.

X error (mean -1.1 cm, std 0.8 cm) is the smallest of the three axes and shows no consistent systematic trend. Values are predominantly small and negative, with no clear dependence on X position or Z distance. The mean is slightly negative and about the same size as the ±1 cm ruler measurement uncertainty, so it is not possible to conclude there is a real lateral bias. At extreme lateral positions (ball near the left or right edge of the camera frame) the error is marginally larger, likely due to reduced stereo matching quality near frame edges, but the effect is small.

Y error (mean -5.0 cm, std 1.1 cm) is the most consistent systematic bias. The camera consistently places the ball 3-7 cm higher than the ruler measurement. The most likely reason is that the pipeline calculates the 3D position from a point on the front surface of the ball, not its centre. Because the camera is mounted above the ball, the closest visible surface of the ball is its top. That point is higher in the image than the true ball centre, so the calculated Y value is smaller (higher in space) than the real position. The larger error at Z = 0.77 m (~5-7 cm) compared to Z = 0.57 m (~3-4 cm) is consistent with this: at greater distance the ball occupies fewer pixels in the image, making the centroid position less precise, and the deprojection formula multiplies the pixel offset by z, so the same pixel-level error produces a larger error in metres. The dataset confirms a stable Y error exists but does not allow us to precisely calculate how much of it comes from this geometry alone.

Z error (mean 4.1 cm, std 0.8 cm) is a consistent positive bias across all captures and distances. The most likely cause is ZED NEURAL depth mode, which tends to overestimate distances at close range (< 1 m). A smaller additional error may come from the ruler reference point: distances were measured from the left camera lens, which may not be exactly the same reference point that the ZED uses internally for depth calculation. The bias does not strongly depend on X or Z

distance within this dataset's range, though there is a slight uptick at extreme lateral positions (ball near the left or right edge of the camera frame) where stereo matching quality is lower. No correction has been applied.

Limitations

- **Ground truth method**: positions were measured by hand with a ruler. Ruler placement error (~1 cm) and ball positioning error contribute to the reported error figures. The error numbers are an upper bound on actual pipeline error.
- **ZED calibration**: the factory calibration is used without any custom recalibration. A custom calibration could reduce the Z bias and any small lens distortion effects contributing to X error at the frame edges.
- **Limited coverage**: the grid does not test Y positions other than floor height, nor Z beyond 0.77 m. Errors may be larger at longer ranges or different heights.
- **Static ball only**: the pipeline is evaluated on a stationary ball. Dynamic captures may introduce additional blur or YOLO confidence effects not captured here.

Recommendations

- **Y correction**: add +5.0 cm to the measured Y value to correct the bias. The std (1.1 cm) is small enough to make this a reliable fix.
- **Z correction**: subtract 4.1 cm from the measured Z value. The low std (0.8 cm) makes this reliable, though positions at the frame edges may need a slightly larger correction.
- **Custom calibration**: if accuracy better than 3 cm is needed, a custom ZED calibration and a correction that varies with distance would be the next step.

Appendix: Per-capture data

Error = camera minus ruler. Blank = out of frame (missed).

#	Image	Det	X (m)	Y (m)	Z (m)	Err X	Err Y	Err Z	|Error|
1	cap_001	yes	0.69	0.22	0.81	-0.01	-0.06	0.04	0.07
2	cap_002	yes	0.59	0.22	0.81	-0.01	-0.06	0.04	0.07
3	cap_003	yes	0.49	0.22	0.81	-0.01	-0.06	0.04	0.07
4	cap_004	yes	0.39	0.22	0.81	-0.01	-0.06	0.04	0.07
5	cap_005	yes	0.29	0.22	0.81	-0.01	-0.06	0.04	0.07
6	cap_006	yes	0.19	0.22	0.81	-0.01	-0.06	0.04	0.07
7	cap_007	yes	0.09	0.23	0.81	-0.01	-0.05	0.04	0.07
8	cap_008	yes	-0.01	0.22	0.81	-0.01	-0.06	0.04	0.07
9	cap_009	yes	-0.12	0.22	0.81	-0.02	-0.06	0.04	0.08
10	cap_010	yes	-0.21	0.22	0.81	-0.01	-0.06	0.04	0.08
11	cap_011	yes	-0.31	0.22	0.81	-0.01	-0.06	0.04	0.08
12	cap_012	yes	-0.41	0.22	0.81	-0.01	-0.06	0.04	0.07
13	cap_013	yes	-0.51	0.22	0.79	-0.01	-0.06	0.02	0.07
14	cap_014	yes	-0.61	0.21	0.81	-0.01	-0.07	0.04	0.08
15	cap_015	yes	-0.73	0.22	0.82	-0.03	-0.06	0.05	0.08
16	cap_016	yes	0.67	0.23	0.74	-0.03	-0.05	0.07	0.09
17	cap_017	yes	0.58	0.23	0.72	-0.02	-0.05	0.05	0.07
18	cap_018	yes	0.49	0.23	0.72	-0.01	-0.05	0.05	0.07
19	cap_019	yes	0.39	0.23	0.71	-0.01	-0.05	0.04	0.07
20	cap_020	yes	0.29	0.23	0.72	-0.01	-0.05	0.05	0.07
21	cap_021	yes	0.19	0.23	0.71	-0.01	-0.05	0.04	0.06
22	cap_022	yes	0.09	0.23	0.71	-0.01	-0.05	0.04	0.06
23	cap_023	yes	-0.00	0.23	0.71	-0.00	-0.05	0.04	0.06
24	cap_024	yes	-0.11	0.23	0.71	-0.01	-0.05	0.04	0.07
25	cap_025	yes	-0.21	0.23	0.71	-0.01	-0.05	0.04	0.07

#	Image	Det	X (m)	Y (m)	Z (m)	Err X	Err Y	Err Z	|Error|
26	cap_026	yes	-0.30	0.23	0.71	-0.00	-0.05	0.04	0.06
27	cap_027	yes	-0.40	0.22	0.70	0.00	-0.06	0.03	0.07
28	cap_028	yes	-0.50	0.22	0.70	0.00	-0.06	0.03	0.07
29	cap_029	yes	-0.61	0.26	0.73	-0.01	-0.02	0.06	0.06
30	cap_030	no	-	-	-	-	-	-	-
31	cap_031	no	-	-	-	-	-	-	-
32	cap_032	yes	0.57	0.24	0.63	-0.03	-0.04	0.06	0.08
33	cap_033	yes	0.48	0.25	0.62	-0.02	-0.03	0.05	0.06
34	cap_034	yes	0.39	0.25	0.61	-0.01	-0.03	0.04	0.06
35	cap_035	yes	0.29	0.25	0.62	-0.01	-0.03	0.05	0.06
36	cap_036	yes	0.19	0.24	0.61	-0.01	-0.04	0.04	0.06
37	cap_037	yes	0.09	0.24	0.61	-0.01	-0.04	0.04	0.06
38	cap_038	yes	0.00	0.24	0.61	0.00	-0.04	0.04	0.05
39	cap_039	yes	-0.10	0.24	0.61	-0.00	-0.04	0.04	0.06
40	cap_040	yes	-0.20	0.24	0.60	0.00	-0.04	0.03	0.05
41	cap_041	yes	-0.30	0.24	0.60	-0.00	-0.04	0.03	0.05
42	cap_042	yes	-0.40	0.24	0.60	-0.00	-0.04	0.03	0.05
43	cap_043	yes	-0.52	0.25	0.62	-0.02	-0.03	0.05	0.06
44	cap_044	no	-	-	-	-	-	-	-
45	cap_045	no	-	-	-	-	-	-	-